Ark–TradeLayer Handshake Spec v0.1

1. Roles

- User (U): TL trader wanting wrapped-ARK-BTC.
- Ark Service Provider (ASP): Runs Ark vMempool and rounds.
- TradeLayer Node (TLN): Verifies TL txs and watches BTC.
- TL■ARK Bridge: Logic that verifies Ark commitments & exits and mints/burns wrapped tokens.

2. Handshake Objects

ArkTemplate:

- version, network
- templateId (SHA256 canonical form)
- internalKey
- leafScripts, leafVersions, leafRoles
- timeouts, semantics
- merkleRoot, taprootOutputKey

ArkRoundDescriptor:

- aspId
- templateId
- roundId
- commitmentTxId
- outputIndex
- outputAmount

3. Handshake Flow

Step 0 – Template discovery:

TL keeps allowlist of templateIds.

Step 1 – Session Init (U ↔ ASP):

User requests templateId; ASP confirms and provides ArkTemplate + deposit address.

Step 2 – Deposit & VTXO assignment:

User deposits BTC. ASP assigns VTXO and provides Merkle branch to user.

Step 3 – TL mint:

User posts ARK_WRAP_MINT TL transaction with:

- templateId
- aspId
- deposit txid:vout
- valueSats
- userVtxoId
- userArkProofHash

TL consensus checks templateId allowlist + on■chain UTXO validity.

4. Redeems (Burn → Ark exit)

Exit spends appear as Taproot script■path spends. TL protocol parses:

- leaf script
- control block
- merkle path
- checks taprootOutputKey reconstructed from leaf matches templateId

TL marks redeem as complete.

5. Protocol■Level Parsing Capabilities

For deposits:

- Parse Taproot outputs
- Check xonly key matches a known template's taprootOutputKey
- Confirm value and confirmation depth

For redeems:

- Parse script■path spends
- Recompute leaf hash → merkle root → tweaked key
- Match leaf script hash to template's "user■exit" leaf

- Verify value release
6. Template Canonicity & Checksums
templateId = SHA256(
TL_ARK_TEMPLATE_V1 ||
network ||
internalKey ||
merkleRoot ||
canonical JSON of leafScripts, versions, timeouts, semantics
)
Protocol requires templateId ∈ ALLOWED_TEMPLATES.
7. Authentication at App Level
- ASP identity aspId (domain or pubkey)
- Signed ArkTemplate and round descriptors
- UI presentation to prevent user mis■routing funds
8. Minimal Parsing Logic
isArkCommitmentUtxo() checks if a Bitcoin UTXO corresponds to known (templateId, aspId).
9. Summary
Handshake layer handles trust, auth, and template disclosure.
Protocol layer verifies:
- canonical template identity
- UTXO adherence to template
- exit leaf spends
Thus enabling safe wrapped token minting and redeeming without exposing users to trap scripts.